

National Research University “Higher School of Economics”.

Faculty of Computer Science. Data Science.

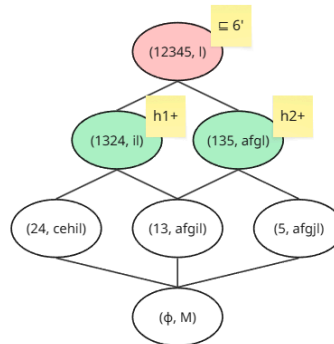
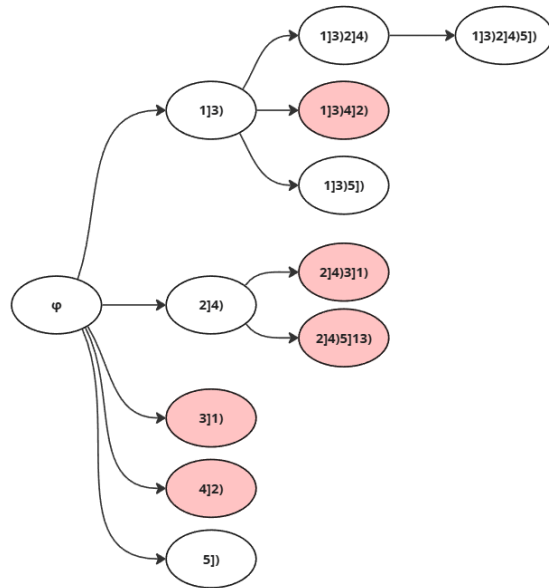
Ordered Sets in Data Analysis.

Homework №2 of Abramov Aleksandr Sergeevich.

Q1

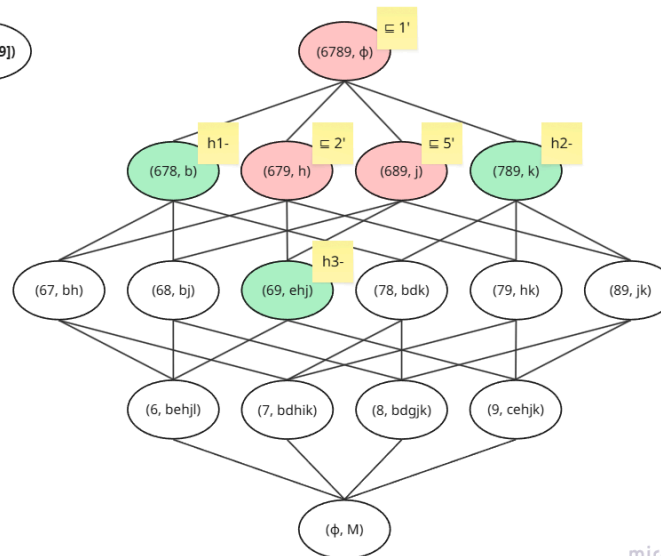
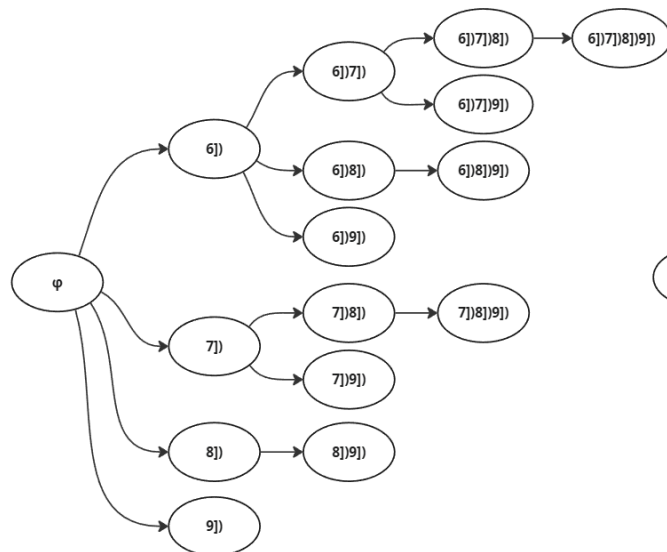
	<i>Grape: Cabernet Sauvignon</i>	<i>Grape: Pinot Noir</i>	<i>Grape: Merlot</i>	<i>Body: Light</i>	<i>Body: Medium</i>	<i>Body: Full</i>	<i>Aging: Oak Barrel</i>	<i>Aging: Stainless Steel</i>	<i>Climate of Region: Warm</i>	<i>Climate of Region: Cool</i>	<i>Alcohol % ≤ 13. 5</i>	<i>Alcohol % ≥ 13. 5</i>	<i>Class</i>
	a	b	c	d	e	f	g	h	i	j	k	l	
1	+					+	+		+			+	+
2			+		+			+	+			+	+
3	+					+	+		+			+	+
4			+		+			+	+			+	+
5	+					+	+			+		+	+
6		+			+			+		+		+	-
7		+		+				+	+		+		-
8		+		+			+			+	+		-
9			+		+			+		+	+		-

Positive hypotheses:



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Negative hypotheses:



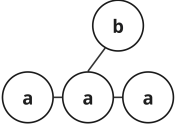
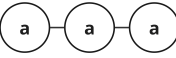
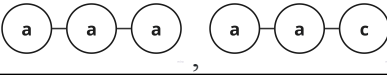
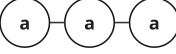
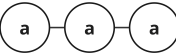
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	a	b	c	d	e	f	g	h	i	j	k	l	
10		+		+				+		+	+		?
11	+					+	+		+			+	?
12		+			+			+	+			+	?
13			+		+		+			+		+	?

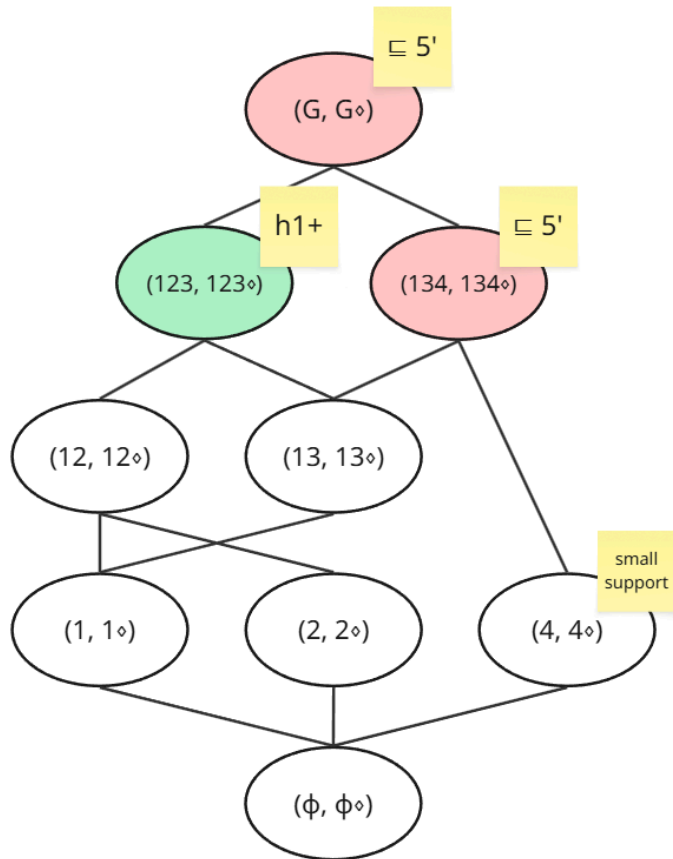
	$h_1^+ = il$	$h_2^+ = afgl$	$h_1^- = b$	$h_2^- = k$	$h_3^- = ehj$	<i>Class</i>
10	No	No	Yes	Yes	No	-
11	Yes	Yes	No	No	No	+
12	Yes	No	Yes	No	No	contradictory
13	No	No	No	No	No	indeterminate

Positive pattern concepts:		
+	$\emptyset \diamond$	
+	$1 \diamond$	
+	$2 \diamond$	
-	$3 \diamond$	
+	$4 \diamond$	
+	$12 \diamond$	
+	$13 \diamond$	
-	$14 \diamond$	
-	$23 \diamond$	
-	$24 \diamond$	
-	$34 \diamond$	

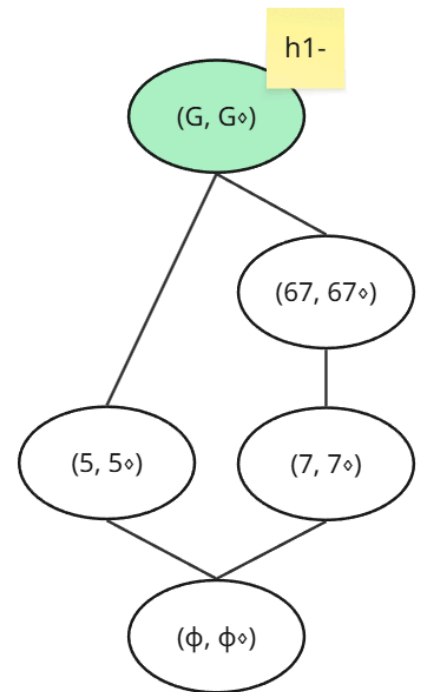
Negative pattern concepts:		
+	$\emptyset \diamond$	
+	$5 \diamond$	
-	$6 \diamond$	
+	$7 \diamond$	
-	$56 \diamond$	
-	$57 \diamond$	
+	$67 \diamond$	
+	$567 \diamond$	

+	123 $\diamond$	
-	124 $\diamond$	
+	134 $\diamond$	
-	234 $\diamond$	
+	G $\diamond$	

Positive hypotheses:



Negative hypotheses:



miro

	h_1^+	h_1^-	Class
8	No	Yes	-
9	Yes	No	+
10	Yes	Yes	contradictory